

LI MOU

Machine Learning Research Engineer | Mobility Data | Multimodal Learning

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Grenoble, France Passeport Talent
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PROFILE

Machine learning and algorithm engineer with experience in large-scale mobility recommendation systems and end-to-end delivery (Spark/Hive/Flink/Kafka, offline evaluation & online experimentation) across global markets. Also building hands-on LLM/RAG and agent-style prototypes for document QA (ingestion, hybrid retrieval, reranking, and prompt iteration) with reproducible code on GitHub.

PROFESSIONAL EXPERIENCE

Machine Learning and Algorithm Engineer

DiDi Global (Overseas Business)

Nov 2024 – Present Beijing, China

- Owned end-to-end applied ML iteration for pick-up points, drop-off points and location naming recommendation across DiDi overseas regions, supporting ~15M daily completed orders; partnered with business owners to define use cases, success metrics and rollout strategy across markets.
- Drove recommender model & feature optimization (Wide & Deep; multi-task learning such as CGC/PLE), and explored generative recommendation feasibility for candidate generation/reranking via offline prototyping and evaluation.
- Built and owned offline feature/training data pipelines using Spark (Scala) + Hive in a Hadoop/HDFS/YARN ecosystem: integrated multi-source data, handled joins/deduplication/late-arriving events/backfills, and produced core data marts/Hive tables with SLA and data quality checks.
- Integrated real-time feature updates via Flink + Kafka by writing key features/states into KV (e.g., Redis) for online inference; connected feature freshness & coverage monitoring (latency, missing rate, drift) and collaborated with backend/platform teams on idempotency and consistency.
- Drove iteration from business pain-point analysis to delivery: proposed feature & model improvements, trained models, and ran offline evaluation via Jenkins pipelines and case-level regression analysis; launched and monitored A/B experiments, analyzed impact, and communicated readouts while supporting SIT/UAT and release coordination with cross-functional stakeholders.
- Drove +0.5pp improvement in key business metrics and reduced case occurrence rate by 0.06pp.

Research Engineer specializing in Machine Learning & Privacy-Preserving Learning

National Institute for Research in Digital Science and Technology (Inria)

Feb 2023 – Apr 2024 Lille, France

- Developed privacy-preserving modules for the TRUMPET project: <https://inria.hal.science/hal-04092216/>
- Designed differential privacy methods for machine learning and database querying, with attention to the balance between privacy guarantees and utility.
- Investigated open-source federated learning frameworks including PySyft, FATE, and Flower, and conducted experiments to evaluate their relevance for the project.
- Developed a survival analysis module in the context of differential privacy and federated learning.

EDUCATION

M.Sc Data Science

Grenoble INP - Ensimag, France

Sep 2020 – Jul 2021

M.Sc Signal and Image Processing Methods and Applications

Grenoble INP - Phelma, France

Sep 2019 – Aug 2020

B.Eng Electrical and Electronic Engineering

National University of Singapore

Aug 2014 – Jul 2018

SKILLS

Machine Learning Deep Learning
Natural Language Processing
Image Processing Data Mining
Privacy Preserving Optimization
Algorithmic Design

Python C/C++ Java Scala
SQL Git Scikit-learn Pytorch
Tensorflow OpenCV Spark
Tornado Linux Latex

LANGUAGES

English French Chinese

REFEREES

Jan Ramon

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Magnet team, Inria Lille

Dr. Sabri Challouf

@ sabri.challouf@gmail.com

Machine Learning Engineer in NLP

Sohu, Inc.

📅 Mar 2022 – Aug 2022

📍 Beijing, China

- Developed NLP models with multi-task learning for article quality assessment across multiple dimensions.
 - Developed an extractive text summarization model for content understanding applications.
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Data Scientist specializing in Recommendation Systems

Baidu, Inc.

📅 Oct 2021 – Feb 2022

📍 Beijing, China

- Developed and optimized large-scale recommendation system tasks for Baidu FEED.
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PROJECTS

Large Language Model Learning Trip (LLM/RAG & Agent Prototyping)

📅 Apr 2024 – Present

- Hands-on learning and prototyping of LLM applications for document QA, focusing on RAG workflows: document ingestion/pre-processing, hybrid retrieval (vector + keyword), reranking, and prompt template iteration driven by failure cases to improve answer accuracy and reduce hallucinations.
 - Built an agent-style conversational QA prototype with LlamaIndex: Query Fusion retrieval, reranking (e.g., bge reranker), question condensation for multi-turn chat, and iterative prompt design; maintained a growing set of golden test cases for regression checks during iteration.
Code: <https://github.com/lokeachirou/LLM/tree/main/Projects/LlamaIndex>
 - Implemented an experimental RAG pipeline with retrieval fusion (RRF), Elasticsearch + vector store, and CrossEncoder reranking; explored chunking strategies and metadata usage for better retrieval on documents with varying structures.
Code: https://github.com/lokeachirou/LLM/tree/main/Projects/rag_and_llm_embeddings
 - GitHub: <https://github.com/lokeachirou/LLM/tree/main/Projects>
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UGA Advanced Learning Model Data Competition

📅 Oct 2020 – Jan 2021

- Designed and demonstrated a kernel-based machine learning algorithm for sequence classification.
 - GitHub: <https://github.com/lokeachirou/Data-challenge-for-kernel-method>
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Data Analysis Based on Spark Database

📅 Nov 2020 – Jan 2021

- Performed distributed data analysis using Spark.
 - GitHub: <https://github.com/lokeachirou/spark-project>
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DISSERTATION WORK

M.Sc Thesis – Natural Language Grounding through Dense Video Captioning

Grenoble Computer Science Laboratory

📅 Feb 2021 – Jul 2021

📍 Grenoble, France

- Studied multimodal learning methods for natural language grounding from video data.

- Demonstrated the effectiveness of Transformer-based and BERT-based models for multimodal problems.
 - Explored the interpretability of multimodal models.
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M.Sc Thesis – Deep Learning-based Phase Retrieval for X-ray Phase Contrast Imaging

CREATIS Bio-imaging Laboratory

📅 Feb 2020 – Aug 2020

📍 Lyon, France

- Implemented nonlinear deep neural networks to address the ill-posed phase retrieval problem in X-ray phase contrast imaging.
 - Strengthened my experience in signal-oriented machine learning and research experimentation.
 - A related paper was submitted to ISBI 2021.
 - GitHub: <https://github.com/lokeaichirou/deep-learning-based-phase-retrieval-for-X-ray-imaging-tomography>
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B.Eng Thesis – Smart Object Counter

Learning and Vision Lab, ECE, NUS

📅 Aug 2017 – Mar 2018

📍 Singapore

- Implemented R-CNN, Fast R-CNN, and Faster R-CNN models for object detection in a real-time vehicle counting system for traffic monitoring.